> FOR SOCIETY

SOUND SOURCE LOCALIZATION

MICROPHONE OPTIMIZATION RESEARCH

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VERSION HISTORY

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INTRODUCTION

The current state of sound localization technology on the PCB (*Printed Circuit Board*) exhibits certain limitations and inefficiencies. To address these challenges effectively, it is imperative to explore avenues for improvement in hardware configurations and setups. This research is used to investigate optimal microphone arrays and their positioning to enhance sound localization accuracy and efficiency.

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1 RESEARCH QUESTIONS

1.1 MAIN-QUESTION

- How can the sound localization be improved with the help of hardware?
- How can the microphones be connected to the teensy 4.1 for the best result?

1.2 SUB-QUESTIONS

- What type of microphone array is the most optimal?
- How far should the microphones be positioned from each other?
- How many microphones can be connected to one teensy 4.1 development board?
- In what manner must the microphones be connected in order for them to operate accordingly?

2 MICROPHONES

When it comes to the microphones, there are currently 4 in use. They are placed in a 2-by-2 array, with an 11cm distance between the microphones. While this setup provides a foundation, it requires refinement to achieve superior outcomes, as suggested by Edwin Oetelaar's previous recommendations for additional microphone mounting points on the PCB.

To facilitate improvement, several approaches can be considered. Firstly, leveraging the existing four microphones while implementing more sophisticated algorithms could enhance performance. Additionally, expanding the number of microphones has shown promise in increasing accuracy, as evidenced in [1], and can be read in the following quote [2]: "Furthermore, it can be seen from Fig. 11 that increasing the number of microphones has no such significant effect on the accuracy of x-direction and z-direction estimates. However, the accuracy of the source estimate in the y-direction is improved by adding more microphones in the four-microphone array geometry. Fig. 11(c).".

2.1 POSITIONING

Regarding to the placement of 4+ microphones, it can be done in various array forms, such as spherical array, linear array and circular array.

- Spherical Array

Microphones are arranged on a spherical surface. This configuration is ideal for capturing sound from all directions and can provide accurate localization in three-dimensional space.

- Linear Array

In the linear array, the microphones are placed in a straight line. The spacing between the microphones can affect the accuracy of localization. A larger spacing can improve accuracy for sources that are far away, while a smaller spacing can improve accuracy for sources that are closer.

- Circular Array

Microphones are arranged in a circle or ring. This configuration can provide better localization performance in all directions compared to linear arrays.

- Planar Array

Microphones are arranged in a two-dimensional plane, such as a square or rectangle. This configuration can provide better localization accuracy in both horizontal and vertical directions.

Considering the 2D nature of the UI used for displaying sound locations, the planar array emerges as the most suitable option, ensuring balanced accuracy and compatibility with the project's current state.

2.2 DISTANCE

Optimal microphone positioning is crucial for achieving desired results. Considering the frequency range of human hearing (20 Hz to 20 kHz), the distance between microphones significantly impacts accuracy. [3] suggests that a 10cm spacing between microphones yields better accuracy compared to an 8cm spacing, although limitations imposed by PCB constraints need consideration.

3 CONCLUSION

In conclusion, optimizing hardware configurations, specifically microphone arrays and their positioning, presents opportunities to enhance sound localization accuracy and efficiency. By exploring various array forms and considering appropriate microphone distances, significant improvements can be achieved, addressing existing pain points and advancing the capabilities of sound localization technology on the PCB.

GLOSSARY

***PCB* – Printed Circuit Board**: A board that connects electronic components using conductive tracks, pads, and other features etched from copper sheets laminated onto a non-conductive substrate. II, 2, 3

REFERENCES

[1] J. Fan, Q. Luo, and D. Ma, “Localization estimation of sound source by microphones array,” *Procedia Engineering*, vol. 7, pp. 312–317, Jan. 2010, doi: 10.1016/j.proeng.2010.11.050.

[2] C. Mahapatra and A. R. Mohanty, “Optimization of number of microphones and microphone spacing using time delay based multilateration approach for explosive sound source localization,” *Applied Acoustics*, vol. 198, p. 108998–108999, Sep. 2022, doi: 10.1016/j.apacoust.2022.108998.

[3] V. de Laar and Versleijen, *Sound Source Localization research*. 2023.